



Cloud Security with AWS IAM



Chouaib youssouf Togola

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": "ec2:*",
7       "Resource": "*",
8       "Condition": {
9         "StringEquals": {
10          "ec2:ResourceTag/Env": "development"
11        }
12      }
13    },
14    {
15      "Effect": "Allow",
16      "Action": "ec2:Describe*",
17      "Resource": "*"
18    },
19    {
20      "Effect": "Deny",
21      "Action": [
22        "ec2:DeleteTags",
23        "ec2:CreateTags"
24      ],
25      "Resource": "*"
26    }
27  ]
28 }
```



Introducing Today's Project!

Project overview

In this project, we will use the IAM feature in AWS to monitor who is authenticated and has access

Tools and concepts

services we used here were the Ec2 instances , the IAM users set up , the IAM user groups creation, the policy creation , the user atlas set up . We learnt how we could share a credential login , by copying the link , or copying and pasting the credentials somewhere safe

Project reflection

this project took less than 2 hours with inconsistencies



Tags

What I did in this step

In this step, I will launch 2 EC2 instances

Understanding tags

Tags are labels and are to reference instances uniquely or used to filter them by group

My tag configuration

The tag I've used on my EC2 instances are called dev and production. The value I've assigned for my instances are network-dev-chouaib



Learn all resources [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

▼ Name and tags [Info](#)

Key [Info](#)

Q Name X

Value [Info](#)

Q network-dev-chi X

Resource types [Info](#)

Select resource ty... ▼

Remove

Instances X

Add new tag

You can add up to 49 more tags.



IAM Policies

What I did in this step

In this step, we are creating an AWS IAM policy that controls what actions a user can perform on EC2 instances. The goal is to give the new intern access only to the development EC2 instance while preventing access to the production instance. This is done by using resource tags (Env = development) inside the policy conditions.

Understanding IAM policies

IAM Policies are set of rules that apply to individual within the instance so as to restrict or give rights , access within certain limits.

The policy I set up

For this project, I've set up a policy using the JSON instead of the visual editor

Policy effect

I've created a policy that...This IAM policy allows a user to fully manage EC2 instances that belong to the development environment, while preventing access to production resources. It also allows the user to view EC2 resources and blocks them from changing tags to avoid unauthorized access escalation.

Understanding Effect, Action, and Resource

the Effect determines whether the policy allows or denies access. Action specifies what operations the user can perform, such as starting or stopping EC2 instances. Resource defines which AWS resources the policy applies to.



My JSON Policy

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": "ec2:*",
7       "Resource": "*",
8       "Condition": {
9         "StringEquals": {
10          "ec2:ResourceTag/Env": "development"
11        }
12      }
13    },
14    {
15      "Effect": "Allow",
16      "Action": "ec2:Describe*",
17      "Resource": "*"
18    },
19    {
20      "Effect": "Deny",
21      "Action": [
22        "ec2:DeleteTags",
23        "ec2:CreateTags"
24      ],
25      "Resource": "*"
26    }
27  ]
28 }
```



Account Alias

What I did in this step

In this step, I will simplify the user login to the AWS account using an Account alias

Understanding account aliases

An account alias is a friendly name unlike bunch of digits that are hard to remember

Setting up my account alias

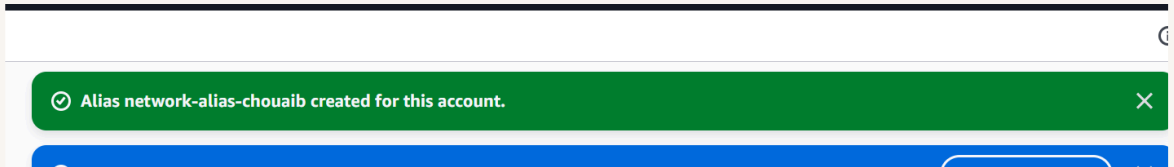
creating it is straightforward . Just go to the IAM > right dashboard > and choose create account Alias



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IAM Users and User Groups

What I did in this step

In this step , we don't want the intern to have access to our atlas account , since we do not want them to have access to the production environment . what we will do : Set up a dedicated IAM group for all NextWork interns, so we can manage all interns' permissions from one place. Set up a dedicated IAM user for the new intern, so he can have a way to log in

Understanding user groups

IAM user groups are collections of people who will have the same access and rights . it is easier to make user groups so as to give them rights at once rather than setting up for each individual which will take a lot

Attaching policies to user groups

I attached the policy I created to this user group, which means they only can achieve what was set as limits within that policy

Understanding IAM users

IAM users are identities in AWS created for people or applications to securely access AWS resources with specific permissions.

Logging in as an IAM User

Sharing sign-in details

The first way is through a .csv file and the second way is by manually copying and sharing the console sign-in URL, username, and password.

Observations from the IAM user dashboard

Once logged in as IAM user, you notice that some of your dashboard panels are showing Access denied already. this is due to the policy we had made.

IAM > IAM users > Create user

✔ User created successfully

You can view and download the user's password and email instructions for signing in to the AWS Management Console.

View user

Step 3

Review and create

Step 4

Retrieve password

Console sign-in details

Email sign-in instructions ↗

Console sign-in URL

https://network-alias-chouaib.signin.aws.amazon.com/console

User name

network-dev-james

Console password

***** Show

Cancel

Download .csv file

Return to users list



Testing IAM Policies

What I did in this step

In this step, we will Log into AWS using the intern's IAM user and test the intern's access to your production and development instance

Testing policy actions

we tested our JSON IAM policy by using the intern account to stop both the dev and prod instances

Stopping the production instance

when trying to stop the instance , a red banner pops . this is because within our policy , the intern is not allowed to stop the instance



⊗

Failed to stop the instance i-06ee7954799cf7119

Diagnose with Amazon Q

×

You are not authorized to perform this operation. User: arn:aws:iam::650539478211:user/network-dev-james is not authorized to perform: ec2:StopInstances on resource: arn:aws:ec2:ca-central-1:650539478211:instance/i-06ee7954799cf7119 because no identity-based policy allows the ec2:StopInstances action. Encoded authorization failure message: KyCyqR4rq-ofelysCkWv1ubYOvp15CZ9kV0XsHUXzkWbTzQZODvWEciUab8o7agwn1oHyr3FCynleaQ1eczK8L3G_qwbm9oz8Xn_t6es9-U1uoFwtudwnV3XaYhO5iDijZFms-Igpr-CjX0yTDiqkMcsiDXA9tclL9P5bdQHREjmw79gri7Vxf0dmKP0oqquWhQKpKo9ODprketFL6mbPgNi6Pa_rvV9-N173h275NP66Cm-F1U7lcvlauCFbb8fCwkVjaeph3RcOPJVbxzobuBgDnglYxh0mvWvQrrMhVVzoRx5Pac7MTa-Dd-XHOsexMJZOmKa2bjMzrYFCxtB9j1Y_4FVMTOKEqeJQrsUMaI5n1mvNe87X95JM_NIWbKd1Dt5n4IMhc8f6arBy8PU4mM3XG0N1HNyO_G9vELPeMMCrEDGJKpxl1iRY8XIBT6y9Ab7pjFAE38tJM7uRKqp7g5MYrxF7_SFDeuQqw8FoL_JW7HKWIZw8qHoxXq4TCJFZmH9BJQow8mz0XcY4mVtEsYCFRwL5fbBP6rXIOIX4-EzADNpxxDu_39z4_u0zGfobHOKYYbS7su8WZX0iLEPILzD9eM_ViWLN48-

i-06ee7954799cf7119 (network-dev-chouaib)

Stopping the development instance

Next, when we tried to stop the development instance, it worked , contrary to the previous. Why ? This was because we had set within our policy that the intern could stop the instance with the development tag , but not the production tag.

⊙ Successfully initiated stopping of i-0d14196fdc9b68c08

Notifications 0 0 4 0 0 0

Instances (1/2) Info

Last updated 9 minutes ago

Connect

Instance state

Actions

Launch instances

Saved filter sets

Choose filter set

Find Instance by attribute or tag (case-sensitive)

< 1 >

⚙

	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone
☒	network-dev-c...	i-0d14196fdc9b68c08	⊖ Stopped	t3.micro	-	ca-central-1b



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